Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

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Step 1 : First , find how far John drives initially .

He drives for 3 hours at 6 0 mph .

Distance = speed ÃĹ time = 6 0 mph ÃĹ 3 hours = 1 8 0 miles .

So , John is 1 8 0 miles away from home when he turns around .

Step 2 : Now , during the return trip , he tries to get home in 4 hours .

He spends the first 2 hours in stand still traffic . So , during that time , he moves 0 miles .

Then , he drives for half an hour (0 . 5 hours) at 3 0 mph .

Distance covered in this period = 3 0 mph ÃĹ 0 . 5 = 1 5 miles .

Then , the remaining time after the first 2 hours and half hour :

Total time allowed = 4 hours .

Used so far : 2 hours (traffic) + 0 . 5 hours = 2 . 5 hours .

Remaining time = 4 - 2 . 5 = 1 . 5 hours .

He drives at 8 0 mph during this time .

Distance covered = 8 0 mph ÃĹ 1 . 5 hours = 1 2 0 miles .

Step 3 : Total distance traveled on the return journey = 1 5 + 1 2 0 = 1 3 5 miles .

Step 4 : So , his distance from home at the end is :

He started 1 8 0 miles away from home , and he drove 1 3 5 miles back toward home .

So , distance from home = 1 8 0 - 1 3 5 = 4 5 miles .

Therefore , John is 4 5 miles away from home at the end .

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<answer> 4 5 </answer>